

Official Agricultural Extension Advisory: Sheath Blight in Rice (Paddy)

Causal Pathogen: Rhizoctonia solani (teleomorph Thanatephorus cucumeris)

Issuing Source: ICAR-Indian Institute of Rice Research (IIRR) Advisory | Date: 2024-02-28

1. Identification and Morphological Symptoms

Initial symptoms develop on leaf sheaths near the water level as oval or oblong, water-soaked, greenish-gray spots (1-3 cm long). As lesions mature, the centers become bleached, grayish-white, with an irregular dark brown or purple border. Under favorable microclimates, lesions coalesce, spread upward to upper leaf sheaths and blades, forming snake-skin-like banded patterns, leading to sheath rot, panicle blighting, and severe plant lodging.

2. Environmental Epidemiology & Causes

Overwinters as sclerotia or mycelium in soil and floating plant debris. Triggered by dense crop canopy, high plant densities, continuous submergence, high relative humidity (85-100%), temperature between 28-32 deg C, and heavy doses of nitrogenous fertilizers.

3. Cultural Management and Field Prevention

Prevention: Opt for wider spacing during transplanting (20 x 15 cm or 20 x 20 cm) to facilitate canopy aeration. Skim off and burn floating sclerotia and stubble during field puddling before final leveling.

Cultural Practices: Ensure alternate wetting and drying (AWD) water management. Avoid excess nitrogen; apply nitrogen in 3 split doses with adequate potash. Keep bunds clean of grassy weed hosts (e.g., Echinochloa colona).

4. Organic and Biological Management

Seed treatment with Trichoderma viride or Pseudomonas fluorescens @ 10 g/kg seed. Soil application of Trichoderma enriched farmyard manure (FYM) @ 2.5 kg/ha mixed with 100 kg FYM at the time of final land preparation. Foliar spray of Pseudomonas fluorescens @ 0.5% (5 g/L) at tillering and boot leaf stages.

5. Verified Chemical Interventions

Foliar spray targeted at the base of the plant: Hexaconazole 5% EC @ 2.0 mL/L or Validamycin 3% L @ 2.0 mL/L or Thifluzamide 24% SC @ 0.75 mL/L or Azoxystrobin 18.2% + Difenconazole 11.4% SC @ 1.0 mL/L at the first appearance of symptoms. Direct spray toward lower sheaths near the water line.

• **Active Ingredient:** Hexaconazole 5% EC | **Reference Formulation:** Contaf (standard reference formulation) | **Registrant:** Rallis India Ltd | **Source:** DPPQS Approved Pesticide List (2024-02-12)

• **Active Ingredient:** Validamycin 3% L | **Reference Formulation:** Sheathmar (standard antibiotic fungicide) | **Registrant:** Dhanuka Agritech Ltd | **Source:** ICAR-IIRR Rice Pathology Bulletins (2024-01-25)

• **Active Ingredient:** Thifluzamide 24% SC | **Reference Formulation:** Pulsor (registered formulation) | **Registrant:** Dow / Gowan / Nissan Chemical | **Source:** CIBRC National Pesticide Gazette (2024-03-05)

Notice: Chemical recommendations are derived strictly from central agricultural research advisories. Farmers must consult local district agricultural extension officers for local dosage and label claims.